Bhuvan Nallamothu

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Internet of Things (Focus: Deep Learning)| GPA: 3.80/4.00 — Aug 2025 - Dec 2026

Northeastern University — Boston, MA
Master of Science in Computer Science| GPA: 4.00/4.00 (Transferred to Carnegie Mellon University) — Aug 2024 - May 2025

SKILLS

Languages: Python, C++, CUDA C/C++, Triton, Go, Java (SpringBoot), SQL, R, MATLAB, Bash, Makefile, Scala
ML Frameworks: PyTorch, TensorFlow, JAX, Hugging Face, NumPy, CUDA Kernels, ONNX Runtime, TensorRT, WandB, PEFT
Gen AI: RLHF (PPO/DPO), LoRA, QLoRA, DeepSpeed ZeRO, SGLang, vLLM, RadixAttention, FlashInfer, VERL, INT4/FP8 Quant
MLOps: Docker, Kubernetes, GitHub Actions, Airflow, Prometheus, Grafana, AWS, GCP, Git, Jenkins, MLflow, DVC, Modal, Slurm
Databases: PostgreSQL, MongoDB, Kafka, Redis, Elasticsearch, Neo4j, Apache Spark, Weaviate, Pinecone, FAISS, pgvector

EXPERIENCE

Autonomy and Perception Engineer — CMU Lunabotics — Jan 2026 - May 2026
- Engineered a real time perception pipeline lifting RGB obstacle detections into 3D LiDAR format point clouds via per pixel depth projection, fusing LiDAR and RGBD into a unified ROS2 costmap for autonomous rover navigation

Lead Teaching Assistant — CMU AI Model Development — Jan 2026 - Mar 2026
- Graded 150+ deliverables across 50 students on production RAG pipeline implementations, evaluating FAISS vector indexing, BM25+vector hybrid retrieval, cross-encoder reranking, multi hop query decomposition, and chunk level citation tracing
- Assessed full research portal deployments for hallucination detection rates, groundedness evaluation metrics, and trust aligned output behavior patterns; verified exportable AI generated artifacts met production grade RAG deployment standards

Research Assistant — JNTU Hyderabad — Mar 2023 - Apr 2024
- Built end to end IoT-ML agriculture decision system: ESP32 firmware with DHT11 and soil moisture sensors over WiFi feeding a Flask API serving ResNet50V2 seed defect classification, Random Forest soil quality (88.89%), and 22-crop recommendations
- Benchmarked 4 transfer learning architectures for agricultural seed defect detection, selecting ResNet50V2 with batch normalization and dropout, integrated with tabular soil models trained on 3 merged IoT and agricultural datasets for full yield prediction pipeline
- Trained a custom CNN on 5,856 chest X-rays with 10 fold Stratified K-Fold CV, class weight balancing, and ReduceLROnPlateau, achieving 96.75% peak and 94.82% mean validation accuracy with Grad-CAM explainability for pneumonia detection

PROJECTS

WorldServe: Autoregressive World Model Inference
- Accelerated Open-Oasis 500M autoregressive diffusion world model inference 3.54x (426.9s to 120.6s) on an H100 SXM, holding coherence within 0.02 dB of the DDIM baseline across 950 frames and 72 configurations over 16 optimization families
- Stacked DPM-Solver++ 2M (Adams-Bashforth-2 multistep, 10 to 5 steps) with an action magnitude difficulty schedule allocating 2, 3, or 5 forwards per frame from the 25 dim action vector, cutting total model forwards 71.8% (9,500 to 2,683)
- Ported TaylorSeer block feature Taylor extrapolation with per frame state reset for a 2.52x standalone speedup at zero validation failures across 152K block calls, the only per step skip method to survive the 950 frame autoregressive loop

CUDA Accelerated LLM Training Engine: Kernels to Transformer
- Implemented CUDA C++ kernel primitives from scratch: stride-based map/zip operators, shared memory tree-reduce for parallel aggregation, and tiled matrix multiplication forming the low level GPU compute backbone of a full LLM training stack
- Assembled miniTorch autograd engine (topological sort DFS backprop, Linear/Dropout/LayerNorm1d) and Pre-LN decoder only Transformer (Multi-Head Attention, causal masking, logsumexp loss), achieving BLEU 20 on IWSLT14 De-En in 10 epochs
- Fused transformer with LightSeq2 inspired fused CUDA kernels: numerically stable softmax (CUB BlockLoad, causal masking) and float4 SIMD LayerNorm, achieving 6.5x/5.5x softmax and 15.8x/3.7x LayerNorm forward/backward speedups

Distributed LLM Training and Inference Optimization (GPT-2, LLaMA-2)
- Developed data-parallel GPT-2 (rank based DataPartitioner, torch.distributed all-reduce, 1.5x+ tokens/sec on 2 GPUs) and pipeline parallel GPT-2 (clock cycle microbatch scheduling, per device worker thread queues), outperforming model parallelism
- Fine tuned LLaMA-2-7B on 2x V100 via DeepSpeed ZeRO sharding and LoRA within 32 GB GPU memory budget, demonstrating parameter efficient adaptation of a 7B-scale model under tight multi-GPU memory constraints
- Deployed SGLang inference with RadixAttention (KV cache prefix reuse via radix tree, LRU leaf eviction, cache aware scheduling prioritizing longer prefix matches) and compressed FSM constrained decoding via FlashInfer backend

RLHF Fine Tuning Pipeline: Reward Modeling and PPO via VERL
- Executed end to end PPO based RLHF via VERL (HybridFlow: decoupled generation/training phases) fine tuning GPT-2, quantifiably shifting reward distribution from -0.5 to $+0.5$ post training, validated against pre RLHF reward baseline
- Constructed DistilBERT reward model with pairwise ranking loss on 10K Anthropic hh-rlhf preference pairs, achieving 60%+ accuracy